MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)

FOURTH SEMESTER EXAMINATION

(2024 ADMISSION ONWARDS)

MG4DSEAQ C203-Seed Production and Hatchery Management of Shellfishes

Duration: 1.5 hrs **Maximum Marks: 50**

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A
Very Short Answer Questions
Answer all 10
Each question carries 1 marks

1. Name any one method used for identifying shellfish seeds. [U] [1]

2. Why is the androgenic gland important in male prawns? [A] [2]

3. Define external fertilization in molluscs. [A] [2]

4. Mention one important selection criterion for crab broodstock. [A] [2]

5. Define nutritional requirements of broodstock. [A] [2]

6. What factors influence the choice of live feed in a shellfish hatchery? [An] [4]

7. Define vibriosis in shellfish larvae. [A] [3]

8. How does a sudden drop in salinity affect shellfish larvae? [An] [5]

9. Suggest one SPS (Sanitary and Phytosanitary) measure to prevent exotic pathogens in larval tanks. [C] [6]

10. Identify a water-quality-related BMP that helps minimize larval mortality. [A][6]

[1x 10=10] Marks

Part B
Short Answer Questions
Answer any 5 Questions out of 8
Each question carries 6 marks

11. Explain different methods of shellfish seed collection and species identification.

[U] [1]

12. Explain the role of reproductive hormones in crustacean and mollusc spawning with examples. [A] [2]

13. Compare lobster and oyster reproduction and its use in seed production. [A] [2]

14. Explain the criteria for choosing lobsters broodstock to ensure high-quality seed production. [A][2]

15. Describe the different methods used for induced maturation in prawns and crabs. [A] [3]

16. Design a shellfish hatchery for prawn, considering site selection and necessary facilities. [S] [6]

17. How to package and transport prawn seeds to ensure their survival? [A][4]

18. Explain live feed culture and use in shellfish hatcheries. [A][4]

[6 x 5 = 30] Marks

Part C
Essay Type Questions
Answer Any One Question
Each question carries 10 marks

19. Explain the life cycle of *Penaeus monodon* and its application in hatchery seed production. [A] [2]

20. Discuss larval diseases in shellfish hatcheries and their management. [A][3]

21. Analyse the water quality parameters in a shellfish hatchery and their impact on larval growth and survival. [An] [2]

[10 x 1 = 10] Marks